

Responsible Machine Learning

Exercise 5

Quantity	LR	GBT
Test AUC	0.735	0.718
Generalization gap (train AUC - test AUC)	-0.008	+0.080
Baseline FPR (African-American / Caucasian)	x0.281 / 0.143	
Baseline AIR	1.961	1.782
AIR under PGD at epsilon = 2.0	1.000	1.782 (attack failed)
Shadow membership inference AUC	0.497	0.500

Part 1. PGD evasion

Under PGD on logistic regression, both group FPRs increase rapidly with epsilon. At epsilon = 0.25, FPR_AA rises from 0.281 to 0.569 and FPR_CA from 0.143 to 0.370. At epsilon = 2.0, both reach 1.000. AIR decreases from 1.961 toward 1.000, but this is not a fairness improvement. It reflects the model classifying every test record as positive. The 4/5ths rule is satisfied only because the model has stopped distinguishing anyone.

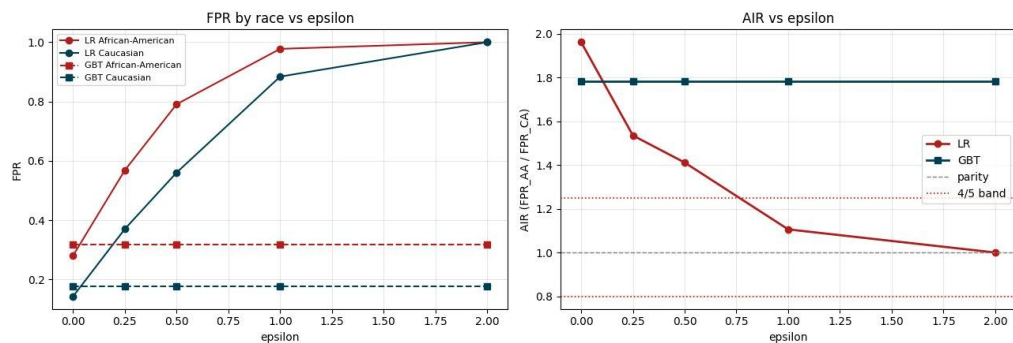


Figure 1. FPR by race (left) and AIR (right) versus PGD perturbation budget for both models.

AIR does not drop below 0.80 for either model in the tested range, since the baseline is already above 1.25. The GBT model shows no change at any tested epsilon, reflecting a limitation of the numerical-gradient attack rather than model robustness: GBT predictions are piecewise-constant in feature space, and small perturbations rarely cross tree split boundaries. A transferability or decision-based attack would be required for a fair comparison; the two models cannot be ranked for adversarial robustness from these results alone.

Part 2. Label-flip poisoning

AUC declines by at most 0.4 percentage points across all tested poison rates and both target groups. By the assignment's stealth criterion (delta AUC at least -2 pp and AIR outside [0.80, 1.25]), all tested rates qualify, but this is an artifact of the baseline AIR already being outside [0.80, 1.25]. The change in AIR is more informative than the threshold crossing. Targeting African-American defendants increases AIR from 1.961 to 3.010 at 30% poisoning, while AUC drops only from 0.735 to 0.731. Targeting Caucasian defendants produces almost no change in AIR (1.961 to 1.940 at 30%).

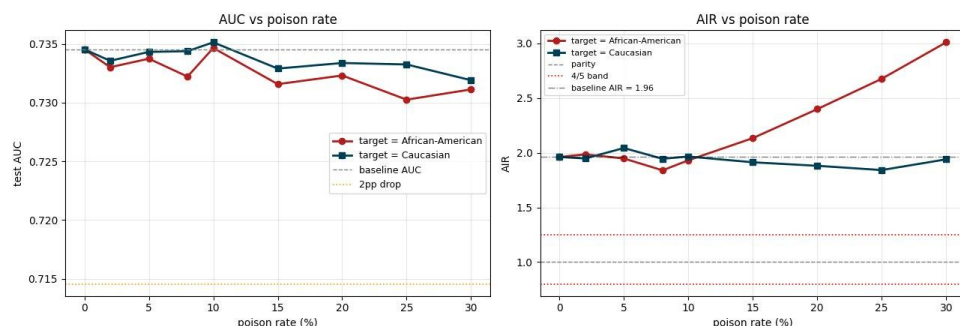


Figure 2. AUC (left) and AIR (right) versus poison rate for both target groups.

The AA-targeted result is counterintuitive: relabeling high-risk African-American records as low-risk increases the AIR ratio rather than decreasing it. Both groups' FPRs decline (FPR_AA from 0.281 to 0.097, FPR_CA from 0.143

to 0.032 at 30%), but FPR_CA falls proportionally faster, widening the ratio. A fairness audit that only tracks the direction of AIR change would conclude that the attack helps African-American defendants, when in fact the disparity has roughly doubled. PSI on input features cannot detect either attack because label-flip leaves the feature distribution unchanged; PSI on output predictions grows with poison rate but only exceeds 0.10 at higher rates.

Part 3. Membership inference

MI AUC is 0.497 for LR and 0.500 for GBT, both close to random. This holds even though GBT shows a generalization gap of +0.080 (train AUC 0.798 versus test AUC 0.718). The confidence gap explains the difference: GBT's mean maximum predicted probability is 0.702 on training members and 0.697 on non-members, a difference of 0.005. The generalization gap is measured in AUC (ranking quality), while the shadow attack exploits absolute confidence values. The two metrics can move independently, and on this dataset the generalization gap does not predict MI vulnerability.

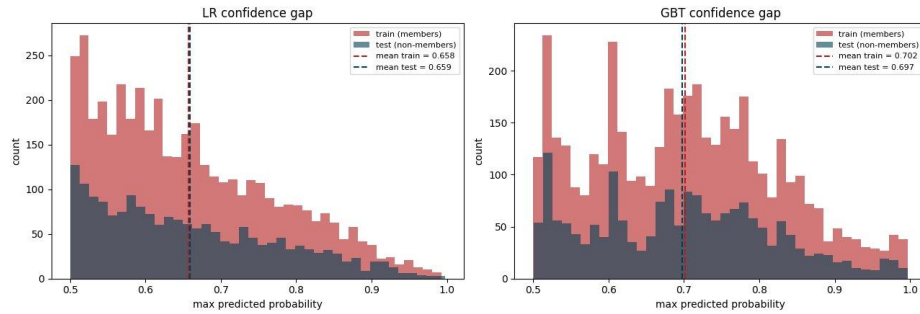


Figure 3. Confidence gap between training members and test non-members for LR and GBT.

The L2 regularization sweep over C in $\{0.01, 0.1, 1.0, 10.0\}$ produces a null result. MI AUC ranges from 0.497 to 0.505, test AUC from 0.732 to 0.735, and AIR from 1.946 to 1.966. The LR model's generalization gap is slightly negative at baseline, so there is no overfitting for L2 to correct. Regularization here neither improves privacy nor reduces accuracy nor affects fairness. This indicates that a default "add L2 regularization" recommendation would not be effective for this pipeline.

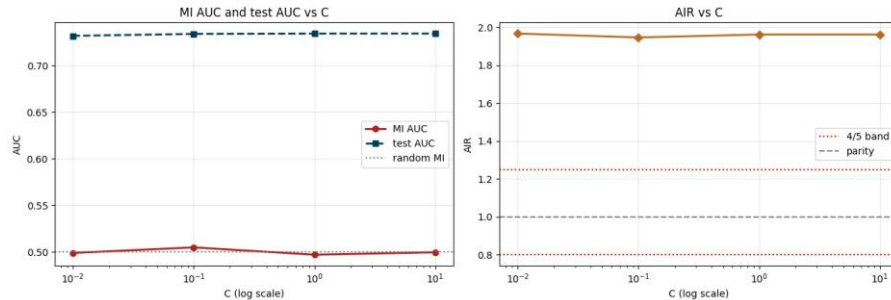


Figure 4. MI AUC and test AUC (left) and AIR (right) versus L2 inverse regularization strength C .

Reflection. Highest-risk finding and mitigations

The highest-risk finding across all three parts is the baseline AIR of 1.961 for the LR model: the unattacked model misclassifies African-American non-recidivists at almost twice the rate of Caucasian non-recidivists. Attack outcomes are secondary to this baseline disparity. PGD inflates both groups' FPRs proportionally, and the membership inference attack is near-random.

Proactive mitigation: an AIR-based deployment gate that requires AIR to fall within $[0.80, 1.25]$ before release. Applied to the experimental results, this gate would block every model tested in the notebook. The four L2 variants had AIR between 1.946 and 1.966, and the poisoned models had AIR between 1.840 and 3.010. The gate is symmetric by definition and triggers in either direction of disparity. Its cost is delayed deployment until the underlying disparity is addressed through threshold calibration, reweighing, or feature changes.

Reactive mitigation: output-distribution PSI monitoring added to the existing input-feature PSI monitor. Feature-PSI is exactly zero under label-flip poisoning at all tested rates, while output-PSI grows monotonically with poison rate and exceeds 0.10 at higher rates for both target groups. This monitor would detect severe poisoning that feature-level PSI cannot see. The mitigation has a known asymmetry: output-PSI is computed globally, so AA-targeted poisoning at 30% produces a larger alarm signal than CA-targeted poisoning at the same rate, because more records are eligible for flipping. The monitor therefore offers stronger early warning for attacks against the larger group, which should be documented at deployment.